

# Inter-Satellite All-Optical Networks (ISAON): Hands-on Demonstration of Orbit-Aware Routing Design and WDM Ground-Emulator Control

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**Abstract** We developed a software prototype for inter-satellite all-optical network operation, demonstrating GUI-based visualisation of orbit-aware routing and live WDM ground-emulator control to monitor 100/200/300 Gbps transmission performance on the path corresponding to Barcelona–Mumbai. ©2026 The Author(s)

## Review

Low Earth Orbit (LEO) satellite optical networks have enabled global, high-speed connectivity in regions underserved by terrestrial fibre infrastructure, through 100-Gbps-class inter-satellite links with coherent receivers and optical amplifiers [1]. However, the continuous growth in traffic and the increasing demand for ultra-low latency in applications such as telemedicine and financial transactions have revealed the limits of architectures that rely on electrical processing within satellites.

To overcome these challenges, the concept of the inter-satellite all-optical network (ISAON) was proposed in [2–9], eliminating conventional optical-electrical-optical (O/E/O) regeneration and switching, reducing redundant processing delay and power consumption. However, unlike terrestrial optical networks, ISAON faces two key challenges: (i) the network topology changes continuously with LEO satellite motion, and (ii) end-to-end transmission performance is difficult to estimate, because multiple optical components with varying characteristics are cascaded in this dynamic environment. To address these issues, we developed an orbit-aware routing framework [5] and a digital twin (DT)-based quality-of-transmission (QoT) estimation technique, enabling efficient network design and operation [6].

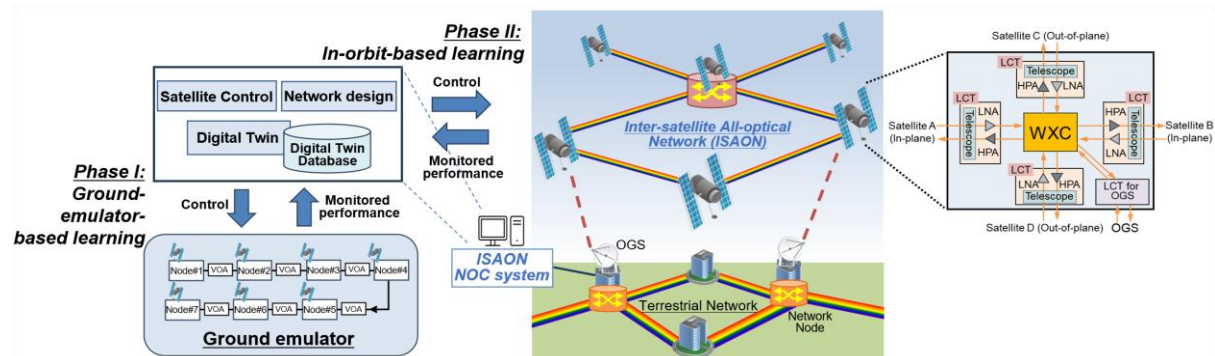
In this paper, we present a live demonstration

using a software prototype of the ISAON network operation centre (NOC) system, showcasing visualisation of orbit-aware routing, live control of the WDM ground emulator located at NICT in Tokyo, Japan, and real-time transmission performance monitoring.

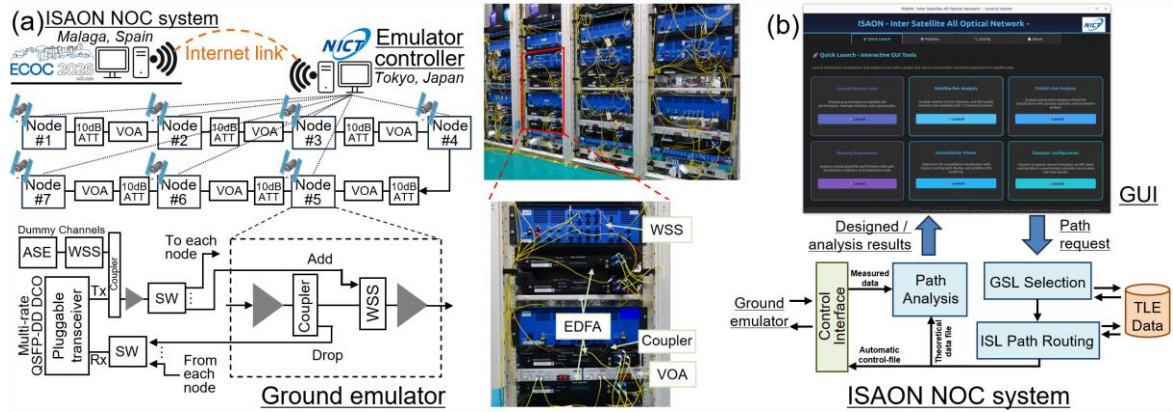
## Novelty

Fig. 1 illustrates the ISAON concept, the NOC system, and the configuration of an all-optical satellite. Each satellite is equipped with laser communication terminals (LCTs) that integrate telescopes, low-noise amplifiers (LNAs), high-power amplifiers (HPAs), a wavelength cross-connect (WXC) switch, and rate-adaptive transceivers, enabling dynamic wavelength-path configuration and switching, analogous to terrestrial optical networks [6–9]. The ISAON NOC system is responsible for route design, satellite control, and overall network management and operation.

As described in the previous section, accurate network design is significantly more challenging for ISAON than for terrestrial networks. To address this, the NOC system incorporates a DT and its associated database (DB), which model the orbital positions of LEO satellites and the inter-satellite transmission performance based on actual QoT measurements. This enables efficient



**Fig. 1:** Concept of the inter-satellite all-optical network (ISAON), its network operation centre (NOC) system, and the configuration of an all-optical satellite. OGS: optical ground station; LCT: laser communication terminal; LNA: low-noise amplifier; HPA: high-power amplifier; VOA: variable optical attenuator. For details of the LCT for OGS, see [6].



**Fig. 2:** (a) Setup schematic and photographs of the ground emulator. ATT: attenuator; ASE: amplified spontaneous emission; WSS: wavelength selective switch. (b) Configuration of the software prototype of the ISAON NOC system and its GUI.

network design and operation, including satellite selection, routing, and transmission capacity allocation, tailored to the dynamic topology and traffic demands of ISAON.

We propose generating and updating the DT and its DB in two phases. *Phase I, ground-emulator-based learning*: prior to deployment, the pre-FEC BER is measured as the QoT indicator using a ground emulator and populated into the DB. *Phase II, in-orbit-based learning*: once ISAON is deployed, idle transceivers monitor the actual QoT of in-orbit inter-satellite links, updating the DB with real measurements. By combining these two phases, the DT evolves continuously with real-world measurements. This enables adaptive, high-fidelity transmission modelling, facilitates optimal routing and data-rate control, and paves the way for a scalable, autonomous ISAON.

### ECOC Relevance

The concept of ISAON, which extends terrestrial optical networking technologies to inter-satellite optical networks, aligns with several ECOC Demo Zone topic categories: novel networking architectures for low-latency applications; autonomous control, orchestration, and telemetry-based network management; and digital twins for optical networks. Through a live demonstration using the developed GUI-based software prototype and our ground emulator, ECOC participants will gain an intuitive understanding of the ISAON concept, its orbit-aware routing framework, and the DT-based QoT estimation approach. We expect that the hands-on interaction and the discussions prompted by this demonstration will contribute to the advancement of the optical communications research community in the non-terrestrial network (NTN) domain.

### Demo Implementation

The objective of this demonstration is to provide ECOC participants with an intuitive, hands-on understanding of the ISAON concept by interactively showcasing the operation of our software

prototype of the ISAON NOC system together with our WDM ground emulator. The key functions implemented in the prototype are: satellite position analysis, OGS–satellite and inter-satellite link (ISL) analysis, routing and path analysis, and emulator configuration and feedback loop. Accordingly, the demonstration is organised in two main parts: orbit-aware routing and its visualisation, and remote control of our WDM ground emulator with the monitored transmission performance, which will be used to generate the DT.

The on-site equipment will consist of a single laptop PC running the ISAON NOC system software prototype (Fig. 2(b)) that will be connected via the Internet to the emulator controller PC, which remains co-located with the WDM ground emulator hardware at our laboratory (Fig. 2(a)).

The WDM ground emulator [6–9], located at NICT in Tokyo, Japan, is shown in Fig. 2(a). It consists of seven nodes with add/drop functionality. Between nodes, a 10-dB attenuator and a variable optical attenuator (VOA) emulate varying inter-satellite distances without introducing fibre dispersion or nonlinear effects. The output power at each node was set to 20 dBm, which is lower than the 30-dBm-class power reported in [2], accounting for additional losses. An OpenZR+-compliant multi-rate QSFP-DD DCO transceiver [10] generates 30/60-Gbaud DP-QPSK (100/200 Gbps) and DP-8/16QAM (300/400 Gbps) signals, which are combined with three ASE-shaped dummy channels to form four 100-GHz-spaced WDM channels [9]. The pre-FEC BER is then measured after transmission.

The demonstration will be presented through the Python-based GUI of the ISAON NOC system. As shown in Fig. 2(b), it comprises six openable panels. The *Routing Design* and *Emulator Configuration* panels are central to this demonstration and are described in detail below. Upon receiving a path request, the system executes orbit-aware routing [5] in two stages: ground-station-to-satellite link (GSL) selection, where each



**Fig. 3:** (a) Orbit-aware routing on software prototype GUI *Routing Design* window for Barcelona–Mumbai path, (b) *Emulator Configuration* window during the running experiment, and (c) theoretical (designed) and monitored transmission performances for Barcelona-Mumbai path from our WDM ground emulator for a single channel at 100/200/300-Gbps data rates

ground station selects the LEO satellite offering the longest line-of-sight communication window; and ISL path construction, where a modified Dijkstra's algorithm builds links in a prioritised sequence: intra-, near-parallel-, and orthogonal-orbit links as a last resort. For this demo, we use two-line element (TLE) data of the Starlink constellation, downloaded on 17 March 2026 [11], and compute the satellite orbits using the Python SGP4 library. The path information is then passed to the path analysis module, which displays satellite data, ground tracks, and transmission distances.

As shown in the *Routing Design* window in Fig. 3(a) for the Barcelona–Mumbai path, four sub-windows display: (I) the selected path, LEO satellite and OGS positions; (II) the total transmission distance with stepwise changes indicating handover to a new set of satellites; (III) ground-to-satellite and inter-satellite information and distances at the time indicated by the red dotted line in (II); and (IV) the condition-file generator, which produces the data file with the theoretical Q-factor values for all rates and WDM channel counts, and the automatic control file (Fig. 2(b)).

The *Emulator Configuration* window shown in Fig. 3(b) connects to the emulator controller PC and sends the automatic control file — containing

VOA settings, data rates, number of WDM channels — through the *Routing Data Transfer* frame, launching the experiment. The GUI first plots the theoretical Q-factor curves; as the experiment runs, the measured pre-FEC BER values (converted to Q-factor) are overlaid in real time for direct comparison with the design, while live operation information from the testbed is displayed in the *Experimental Results Monitor* frame. Fig. 3(c) shows the monitored transmission performance of the Barcelona–Mumbai case, for a single channel at 100/200/300 Gbps from the WDM emulator. The 400-Gbps case falls below the FEC threshold for the assumed 5-dB amplifier noise figure (NF) and is omitted.

Attendees will interact directly with the GUI: (a) select the OGS pair and routing algorithm, observing in sub-windows (I)–(III) how the path, satellite handovers, and link distances evolve over time; (b) adjust the condition-file parameters in sub-window (IV) (e.g. NF, data rates, WDM channel count); and (c) launch the experiment and watch the measured Q-factor values appear live over the theoretical curves. These interactions show how the software prototype bridges routing design, automated orchestration, and experimental validation, representing an important step toward the practical deployment of ISAON.

## Acknowledgements

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